Project Planning & Management

Intelligent Payment Fraud Detection and Risk-Based Decision System

Data Science and Artificial Intelligence Track

1 Project Overview

Project Name: Intelligent Payment Fraud Detection and Risk-Based Decision System

Project Track: Data Science and Artificial Intelligence

The project aims to develop a Machine Learning-based system that detects potentially fraudulent financial transactions, assigns risk scores, and explains the main factors influencing its predictions. The system will also support risk-based decisions by prioritizing suspicious transactions for further investigation.

The project will use the IEEE-CIS Fraud Detection dataset from Kaggle and deliver a trained Machine Learning model alongside an interactive dashboard.

2 Problem Statement

Financial fraud detection is challenging because fraudulent transactions represent a small proportion of total transactions. This class imbalance can lead to misleading accuracy scores and poor fraud detection performance. Additionally, false alarms and missed fraudulent transactions can result in financial losses and unnecessary investigation costs.

The project addresses these challenges by combining fraud detection, explainable predictions, and risk-based decision support.

3 Project Objectives

- Analyze and preprocess financial transaction data.
- Develop and compare Machine Learning models for fraud detection.
- Evaluate performance using appropriate metrics for imbalanced data.
- Generate risk scores and explain model predictions.
- Build an interactive dashboard to visualize suspicious transactions and model results.
- Analyze the trade-off between detecting fraud, false alarms, and missed cases.

4 Project Scope

In Scope:

- Data preprocessing and Exploratory Data Analysis (EDA).

- Machine Learning model development and comparison.
- Model evaluation using Precision, Recall, F1-score, and PR-AUC.
- Risk scoring, model explainability, and decision-cost analysis.
- Interactive dashboard development and system testing.

Out of Scope:

- Direct integration with real banking systems.
- Automatically blocking real transactions.
- Guaranteeing fraud detection or claiming validated performance on Egyptian banking data.

The project will be an academic prototype and decision-support tool.

5 Project Methodology

The project will follow these main stages:

1. Dataset exploration and literature review.
2. Data cleaning, preprocessing, and feature engineering.
3. Exploratory data analysis.
4. Baseline model development and comparison with alternative models.
5. Model evaluation and improvement.
6. Risk scoring and prediction explainability.
7. Dashboard development and integration.
8. Testing, documentation, and final presentation.

6 Project Timeline

The following is a proposed eight-week plan, subject to the official project schedule.

Week	Activities
1	Planning, dataset exploration, and literature review
2	Data preprocessing and EDA
3	Baseline model development
4	Model comparison and class imbalance handling
5	Model evaluation and improvement
6	Explainability and risk-based decision support
7	Dashboard development and integration
8	Testing, documentation, and presentation

7 Team Roles and Responsibilities

The four team members will share responsibilities as follows:

- **Member 1 — Data Analysis:** Data exploration, cleaning, preprocessing, and EDA.
- **Member 2 — Machine Learning:** Model development, training, and comparison.
- **Member 3 — Evaluation and Explainability:** Performance metrics, model interpretation, and decision-cost analysis.
- **Member 4 — Dashboard and Integration:** Dashboard development, visualization, and system integration.

All members will contribute to documentation, testing, and the final presentation.

8 Risk Management

Risk	Mitigation
Class imbalance	Use appropriate metrics and test suitable balancing techniques.
Overfitting or data leakage	Apply proper validation and data-splitting procedures.
High false positive rate	Analyze Precision, false alarms, and decision thresholds.
Missed fraudulent transactions	Monitor Recall and investigate false negatives.
Limited computational resources	Start with baseline models before testing more complex approaches.
Limited applicability to Egypt	Document dataset limitations and avoid unsupported claims about local performance.

9 Success Criteria and KPIs

Project success will be measured using:

- **Model Performance:** Precision, Recall, F1-score, PR-AUC, and False Positive Rate.
- **Decision Support:** Comparison of detected fraud, missed cases, and estimated decision costs.
- **System Functionality:** Successful predictions, risk scoring, explainability, and dashboard visualization.
- **Project Delivery:** Completion of the prototype, testing, documentation, and presentation.

Numerical performance targets will be determined after initial experiments, and results will only be reported after testing.

10 Expected Deliverables

The expected deliverables include:

1. A data preprocessing pipeline.
2. A trained and evaluated fraud detection model.
3. A model evaluation report.
4. A risk-scoring and decision-support component.
5. An interactive dashboard.
6. A shared GitHub repository containing the source code and documentation.

11 Conclusion

The project combines Machine Learning-based fraud detection with explainable risk assessment and decision support. The final prototype will demonstrate how suspicious transactions can be identified, evaluated, and prioritized for review while highlighting the limitations of the available data and model predictions.